Modelling Pendulums

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Jhondeff Fadullo

1 Introduction

This paper is my results and methods of simulations regarding pendulums, the language of C++ will be dear to us as we proceed through each section. I will first give a review of materials that we must know in order to model such phenomenas. More specifically, we will start with Hooke's law, which will introduce us to this notion of Simple Harmonic Motion. I wrote this paper originally as a way to gather my thoughts and build materials that I needed to model a double pendulum, which means this is more informal that it is formal, but I will try my best to use certain language to promote generality of terms, I also wanted to make it reproduceable, so you, the reader, can follow along.

1.1 Prerequisites

We should get this out of the way, you are going to need certain tools on your device (performed on Windows, but can be performed on other operating systems as long as you have the same tools)

1. A modern C++ compiler supporting at least the C++17 standard (such as `g++`, `clang++`, or MSVC).

2. A terminal or command-line interface to compile and execute the source files.

2 Theory

Starting off with the infamous single pendulum, we can use polar coordinates to simplify this example. The bob of mass makes an angle ϕ with the vertical axis, since we are using polar coordinates, we have the unit vectors $\hat{\phi}$ that tracks the bob of mass tangential motion, while $\hat{\mathrm{r}}$ follows the direction of the pendulum's length.